# Logistic Regression and Classification

*Week 5 – Day 1 | Python and Machine Learning*

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## 1. Logistic Regression

Logistic Regression is a supervised machine learning method. It is mostly used when we need to predict a category, such as yes or no. It learns from examples where the correct answer is already known.

Some uses are:

- Checking whether a bank transaction is fraud or genuine
- Predicting whether a customer may leave a service
- Classifying a message as spam or normal

## 2. Classification

Classification means predicting a group or label instead of a continuous value. In a two-class problem, labels are often written as 0 and 1. For example, 0 may mean genuine and 1 may mean fraud. The meaning depends on the dataset.

## 3. Probability

Probability is a value between 0 and 1 that represents how likely something is. A value near 0 means less likely, while a value near 1 means more likely. For example, a predicted probability of 0.85 means the model estimates an 85% chance. It is an estimate, not a guarantee.

## 4. How the model makes a prediction

The model takes input details called features. It learns the connection between those features and the target label during training. It calculates a score and converts that score into a probability.

## 5. Sigmoid function

The sigmoid function converts a score into a value between 0 and 1. This is useful when the model needs to estimate the probability of class 1.

p = 1 / (1 + e■■)

Here, z is the score produced by the model and e is a mathematical constant. If z is 0, the sigmoid value is 0.5.

## 6. Decision threshold

The probability is converted into a class using a threshold. A common threshold is 0.5. If the probability is 0.5 or more, the model predicts class 1. If it is below 0.5, it predicts class 0. The threshold can be adjusted depending on the problem.

## 7. Python example

In this example, the model predicts whether a transaction is genuine (0) or fraud (1), using a simple feature value. The small dataset is only for practice.

```
from sklearn.linear_model import LogisticRegression

X = [[1], [2], [3], [4], [5], [6]]
y = [0, 0, 0, 1, 1, 1]

model = LogisticRegression()
```

```
model.fit(X, y)

new_value = [[4.5]]
prediction = model.predict(new_value)
probability = model.predict_proba(new_value)

print("Predicted class:", prediction[0])
print("Class probabilities:", probability)
```

## 8. Meaning of the main code lines

- X – stores the input feature values.
- y – stores the correct class for each training example.
- LogisticRegression() – creates the model.
- fit(X, y) – trains the model using the examples.
- predict() – returns the predicted class.
- predict_proba() – returns the probability for each class.

## 9. Reading the output

The predict_proba() result contains two probability values in the order of the model classes. For example, [0.30, 0.70] means 30% for class 0 and 70% for class 1. The model would predict class 1 in this example. Your actual values may vary.

## 10. Regression and classification

## 11. Quick practice

1. A model gives a probability of 0.82 for class 1. What class is predicted with a threshold of 0.5?

Answer: Class 1, because 0.82 is above 0.5.

2. A model gives a probability of 0.25 for class 1. What class is predicted with a threshold of 0.5?

Answer: Class 0, because 0.25 is below 0.5.

## What I learned

I learned that Logistic Regression is used for classification. It first calculates a probability and then uses a threshold to decide the class. I also learned the basic use of fit(), predict() and predict_proba() in Scikit-learn.

| Point | Regression | Classification |
|---|---|---|
| Output | A numerical value | A category or label |
| Example | Predicting house price | Fraud or genuine |
| Common method | Linear Regression | Logistic Regression |